

Cumulative Reflection

As I near the completion of my degree in Cybersecurity Engineering at Iowa State University, I find myself reflecting on the small moments, experiences, and challenges that have shaped me as an engineer. I came to Iowa State unsure if this really was the best decision, but I could not have anticipated how much the university's people and programs would influence my growth.

I have been overly hard on myself to produce good work, and that did not change during my time at Iowa State, but my perspective of what is *good* changed. My early coursework in maths, sciences, and introductory programming courses built on the predictable rhythm: follow the process, avoid mistakes, arrive at the expected result. CYBE 2310, my second cybersecurity course, broke that trend. I was forced to look at problems from a new perspective, to think like a hacker. I had to stop looking for one correct answer and start looking for the answer no one had considered. I wasn't following a procedure; I was learning to understand the system and building a mindset that I would take with me beyond my degree. That shift, from executing steps to thinking like the adversary, was the first sign that good work was going to mean something different.

During my first year at Iowa State, I also took chances that I now realize have changed my educational and professional career forever. I received a one-off email from an assistant of a cybersecurity professor on campus, looking for a new lab member for a team called ISEAGE. The opportunity seemed like a far reach for my skill set, and I had never competed in what they called a *Cyber Defense Competition* before. I reviewed it with my peer mentor at the time and decided to submit my resume and hope for the best with what limited security skills I already had. Sure enough, I was interviewed by a small group of ISU students and received the role.

I was thrown right into the deep end: building virtual environments, implementing vulnerabilities, creating scripts and applications, and, oddly enough, working with people, as a team and as part of an organization running a competition. One of my first projects was building an X-ray machine application and the web server that supported it. I had never used Apache before and had never developed a full-stack application. There was no step-by-step process for me to follow or example that I could simply recreate. I had to teach myself unfamiliar technologies, figure out how the different pieces of the application communicated, troubleshoot problems as they arose, and keep experimenting until I understood how everything worked together.

I had limited knowledge in all of these new areas, but that was a huge step forward in my learning process. I began to realize that learning cybersecurity was not something I would perfect in the classroom. Through ISEAGE, I could apply the knowledge I had built in class, experiment and break things, troubleshoot, and learn from problems that did not already have an expected answer. I began to understand that engineering was not always about knowing how to build something before I started. Sometimes it was about being willing to start with what I knew, identify what I did not know, and teach myself what I needed to solve the problem.

As I gained more technical experience through ISEAGE, I began to see that cybersecurity was about more than the systems themselves. The systems I was building were ultimately being used by other people, and their experience with those systems mattered. I would encounter this idea from a different perspective when I began working in IT for a city government. There, I was no longer primarily building the systems that people interacted with; I was helping the people interact with them.

I supported users in a variety of ways, from access control to printer support. While sometimes I was working on underlying technical projects, most of my job was to support the person on the other side of the screen. Building relationships with the people I supported helped me perform my job better. The more I understood

their roles, workflows, and needs, the better I could understand the problems they were experiencing and find solutions that actually worked for them. I also learned that sometimes the biggest challenge is not solving the technical problem but getting someone to ask for help in the first place. People can be hesitant to admit that they do not understand something, that they made a mistake, or that something is not working. Building trust made it easier for people to reach out before a small problem became a larger one. I learned that IT is not a desk job; it is a people job. The technical solution matters, but so does the trust and relationship that allow you to understand the problem in the first place.

My time studying abroad at the University of Limerick also changed how I viewed my education outside of the immediate environment of Iowa State. Being immersed in another country gave me the opportunity to see how people approach education, technology, and daily life differently. It reminded me that the systems I work with are not isolated to one organization or country. Cybersecurity is inherently global, and the decisions made by engineers can affect people across borders.

I have carried these lessons with me as a student, a TA, and now my corporate security internship. At Principal I have received even more opportunities to innovate and impact a company on a larger scale as a data security intern. There, I can apply more of the technical knowledge that I've learned in class and expand it to new tools and teams. By utilizing the connections I have built through Iowa State, with students, staff, and alumni, I have been able to explore security roles beyond my original team placement. This has helped me immensely with helping me be intentional about the areas of cybersecurity I want to pursue full-time and develop relationships that will continue to grow beyond graduation.

I have truly valued my education at Iowa State, and I can confidently say that I would not be the student, engineer, or leader I am today without the relationships and opportunities I have found here. When I first chose this university, I was unsure if this was the right decision for me. I understand now that the answer was not something I could have known on my first day. I had to experience it through the people I met, the opportunities I took, and the challenges that pushed me beyond what I thought I could do. I am leaving grateful that I gave myself the chance to find out all that I can be.